Materials and Design 45 (2013) 616–627

Contents lists available at SciVerse ScienceDirect

Materials and Design

journal homepage: www.elsevier.com/locate/matdes

Technical Report

Development of constitutive models for dynamic strain aging regime in Austenitic stainless steel 304

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ARTICLE INFO

Article history:
Received 18 June 2012
Accepted 23 September 2012
Available online 2 October 2012

ABSTRACT

The experimental stress–strain data from isothermal tensile tests over a wide range of temperatures (623–923 K at an interval of 50 K), strains (0.02–0.30 at an interval of 0.02) and strain rates (0.0001, 0.001, 0.01, 0.1 s^{-1}) were employed to determine the Dynamic Strain Aging (DSA) regime and to formulate a suitable constitutive model to predict the elevated-temperature deformation behavior in DSA regime of Austenitic Stainless Steel (ASS) 304. Four models, namely, Johnson Cook (JC) model, modified Zerilli–Armstrong (m-ZA) model, modified Arrhenius type equations (m-Arr) and Artificial Neural Networks (ANNs), were investigated. Suitability of these models was evaluated by comparing the correlation coefficient, average absolute error and its standard deviation. It was observed that JC, m-ZA and m-Arr model could not effectively predict flow stress behavior of ASS304 in DSA regime, while the predictions by ANN model are found to be in good agreement with the experimental data.

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1. Introduction

Austenitic Stainless Steel (ASS) 304 has found various applications in the field of defense and nuclear science because of its excellent corrosion resistance in seawater environment due to having addition of molybdenum which prevents chloride corrosion. It has low carbon content due to which it has improved wear and friction properties, less carbide precipitation in the heat-affected zone during welding and a lower susceptibility to intergranular corrosion [1]. This steel is particularly useful in nuclear reactors as it is used for the purpose of cladding of fuel rods. As the temperature attained in the reactors is very high, it becomes essential to study the properties of the material at elevated temperatures. Generally, these steels exhibit a peculiar phenomenon at elevated temperatures with low strain rates in which the mobility of solute atoms can become large enough that they can follow a dislocation during its motion and segregate to its core while it has to wait in front of an obstacle. The repeated segregation and detachment process causes the flow stress to oscillate. In such cases, serrations are observed, i.e. wavy pattern like saw tooth in the stress–strain plot. This effect is called Dynamic Strain Aging (DSA) or Portevin-Le Chatelier effect. Also, DSA is manifested by a negative strain rate sensitivity, which results in unstable, jerky flow [2,3].

Constitutive equations give a mathematical representation of the flow stress behavior of materials and it involves a particular number of material constants which are evaluated using limited number of experimental data [4]. These equations are used in finite element software to simulate the material's response under specified loading conditions [5]. Therefore, the accuracy of the numerical simulation largely depends on how accurately the deformation behavior of the material is being represented by the constitutive equation. Recently, several constitutive models have been developed which can be broadly classified into three categories, namely phenomenological constitutive models, physical based constitutive models and artificial neural networks [6]. In this work, Johnson Cook (JC), modified Zerilli–Armstrong (m-ZA), modified Arrhenius type equations (m-Arr) and Artificial Neural Networks (ANNs) model are employed on the flow stress data of ASS304. Out of these models, JC and m-Arr models are phenomenological constitutive models, while m-ZA model is a physical based constitutive model.

The JC model as put forth by Johnson and Cook [7] has been successfully used for a variety of materials for different ranges of temperature and strain rate as it requires fewer material constants and also few experiments to evaluate these constants. A number of modifications to this model are also available in literature. To represent the behavior of the materials with different strain rate sensitivity and to incorporate adiabatic temperature rise during material deformation, the model has been revised [8,9].

On the other hand, the m-ZA model formulated by Zerilli and Armstrong [10] model has been used for different FCC and BCC

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http://dx.doi.org/10.1016/j.matdes.2012.09.041

Table 1
Chemical composition of Austenitic stainless steel 304 (wt.%).

Element	Fe	C	Mn	Si	Mo	Co	Cr	Cu	Ni	Others
Composition (%)	70.780	0.025	1.140	0.410	0.360	0.210	18.400	0.180	8.190	0.305

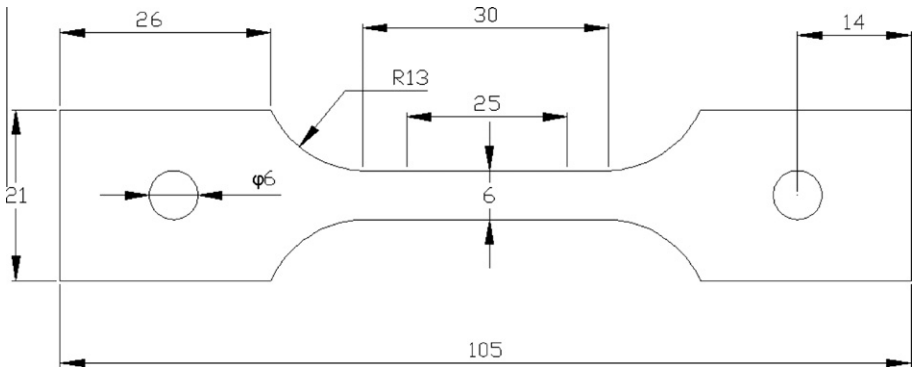


Fig. 1. Flat UTM specimen of 0.9 mm thickness used for tensile testing (in mm).

materials over different strain rates and at temperatures between room temperature and $0.6T_m$, where T_m is the melting point of the material [11]. The m-ZA model is preferred over JC model as it considers the coupled effects of strain rate and temperature. However, it is particularly not suitable to predict the flow behavior of material at temperatures above $0.6T_m$ and at lower strain rates by this model [8,9].

The sine-hyperbolic law in an Arrhenius-type equation has been successfully applied for prediction of high-temperature flow behavior of materials. The relation between the flow stress, temperature and strain rate, particularly at elevated temperatures, can be expressed by an Arrhenius-type equation [12]. The original model has been revised several times to suitably represent the high-temperature flow behavior of various grades of materials. In this model, an exponential strain-dependent parameter was introduced in the sine-hyperbolic constitutive equation to predict the flow stress. The combined effect of the temperature and strain rate on the deformation behavior is represented by the Zener–Hollomon parameter (Z) in an exponent-type equation [13].

Recently, ANN has been applied for describing the hot deformation processes [14,15]. ANN is an artificial intelligence technique that emulates the behavior of biological neural systems in digital software or hardware and this approach need not to have a well-defined process for algorithmically converting an input to an output. ANN methods are very valuable in processes where a complete understanding of the physical mechanisms is very difficult, or even impossible to acquire and where no satisfactory analytical model exists. The greatest advantage of ANNs is their ability to be used as an arbitrary function approximation mechanism that 'learns' from a collection of representative data of the desired mapping. The ANN then adapts itself to reproduce the desired output when presented with training sample input. ANN is ideally suited for the problem of estimating the flow stress from the available experimental data because of its inherently high parallelism. This model does not require explicit mathematical and physical knowledge of deformation mechanism and has good generalization performance. As compared to modeling by mathematical equations, the understanding of flow stress behavior in DSA regime becomes easier by using ANN modeling. [16].

The objective of this paper is to make a comparative study on JC, m-ZA, m-Arr and ANN models on their capability to represent the flow behavior of ASS304 at elevated temperatures and also to capture the effect of DSA phenomenon. Experimental data from isothermal uniaxial tensile tests at elevated temperatures were employed to determine the material constants of these models. Subsequently, the suitability of these models were evaluated by comparing the correlation coefficient, the number of material constants involved and the ability to track the deformation behavior.

2. Experimental details

The composition of ASS304 is given in Table 1. Flat tensile test specimens of thickness 0.9 mm were used to perform the experiments. The dimensions of the specimen were as per the Defence Metallurgical Research Laboratory, Hyderabad (DMRL) standards as shown in Fig. 1, which is a reduced version of ASTM: E8/E8M-11 standards. The samples were machined out of the raw sheet material by wire-cutting electro-discharge machining process for high accuracy and finish. Isothermal tensile tests were conducted on a computerized UTM having maximum load capacity of 100 kN in order to determine the flow stress behavior as well as to study the dynamic strain aging phenomenon. The UTM is equipped with a controlled system to impose exponential increase

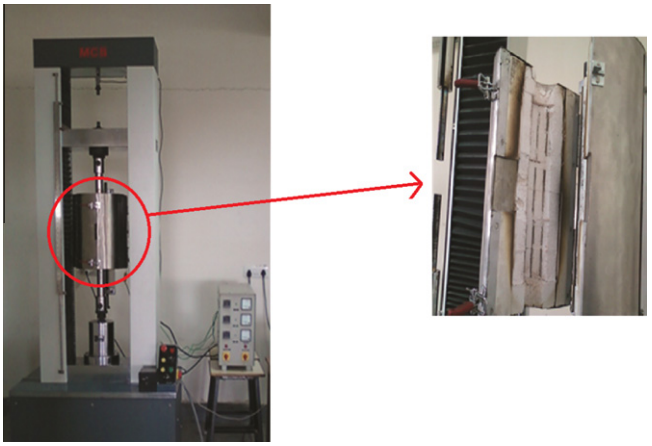


Fig. 2. Computerized UTM of 100 kN capacity with resistance heating 3-zone split furnace.

Table 2
Temperatures and strain rates of DSA regime for Austenitic stainless steel 304.

Temperature (K)	723, 773, 823, 873, 923
Strain rates (s^{-1})	0.0001, 0.001, 0.01

of the actuator speed to obtain constant true strain rates. As shown in Fig. 2, the UTM is also equipped with a resistance heating 3-zone split furnace to conduct the tensile tests at elevated temperatures up to 1000 °C. The pull rods of UTM for the elevated temperature testing are made of Nickel base super alloy CM-247.

The experiments were conducted at four different strain rates (0.0001, 0.001, 0.01 and 0.1 s^{-1}) and various temperatures ranging from 623 K to 923 K at an interval of 50 K. A computer control system is used to record the load vs. extension data, where extension was measured using a high temperature contact-type extensometer. The load vs. extension data was converted into true stress and true plastic strain data. Among all the conducted experiments, the strain rates and temperatures at which DSA phenomenon is observed are presented in Table 2. For data analysis, the true plastic strains are taken from 0.02 to 0.30 at an interval of 0.02.

3. Development of constitutive models

In this work, four-different constitutive models are considered: JC, m-ZA, m-Arr and ANN model. For flow behavior at elevated temperatures, JC model considers isotropic strain hardening, strain rate hardening and thermal softening as three independent phenomena, while m-ZA model also considers the coupled effects of temperature and strain rate on the flow stress. In m-Arr model, the effects of deformation temperature and strain rate are represented by the Zener–Holloman parameter and also the effect of strain on the deformation behavior was incorporated using an

Table 3
Material constants for JC model.

Parameter	A (MPa)	B (MPa)	n	C	m
Value	209.696	1383.2	0.9151	−0.0095	0.5147

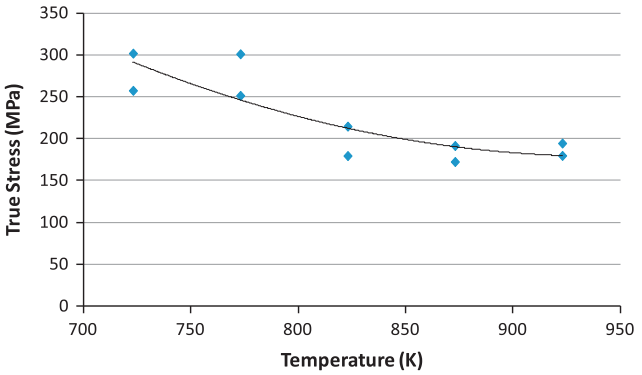


Fig. 3. Plot of true stress and temperature at 0.0001 s^{-1} strain rate for modified ZA model.

Table 4
Material constants for modified ZA model.

Parameter	C_1 (MPa)	C_2 (MPa)	C_3	C_4	C_5	C_6	n
Value	209.696	1463.1	0.000475	0.0049	−0.0386	0.0010	0.9231

Table 5
Material constants for modified Arrhenius model.

Parameter	α (MPa^{-1})	n	Q (kJ/mol)	A (s^{-1})
Value	0.0014969	18.6278	290.64	2.0173×10^{18}

Table 6
β_1 Values.

$\dot{\varepsilon}/s^{-1}$	723 K	773 K	823 K	873 K	923 K
0.0001	1.650259	1.760233	1.113935	0.833389	1.33261
0.001	1.419633	1.723177	1.165669	0.81459	1.435414
0.01	1.524925	1.684552	1.625789	1.075955	1.424423

Table 7
β_2 Values.

$\dot{\varepsilon}/s^{-1}$	723 K	773 K	823 K	873 K	923 K
0.0001	0.325019	0.307148	0.223002	0.130431	0.221308
0.001	0.325214	0.341067	0.258268	0.183777	0.158067
0.01	0.373485	0.353261	0.331759	0.20444	0.217855

Table 8
β_3 Values.

$\dot{\varepsilon}/s^{-1}$	723 K	773 K	823 K	873 K	923 K
0.0001	−0.56754	−0.91291	0.212577	0.200978	0.466884
0.001	−0.47449	−0.21617	−0.19424	−0.28956	−0.55051
0.01	0.017327	−0.08688	0.01694	−0.66627	−0.32372

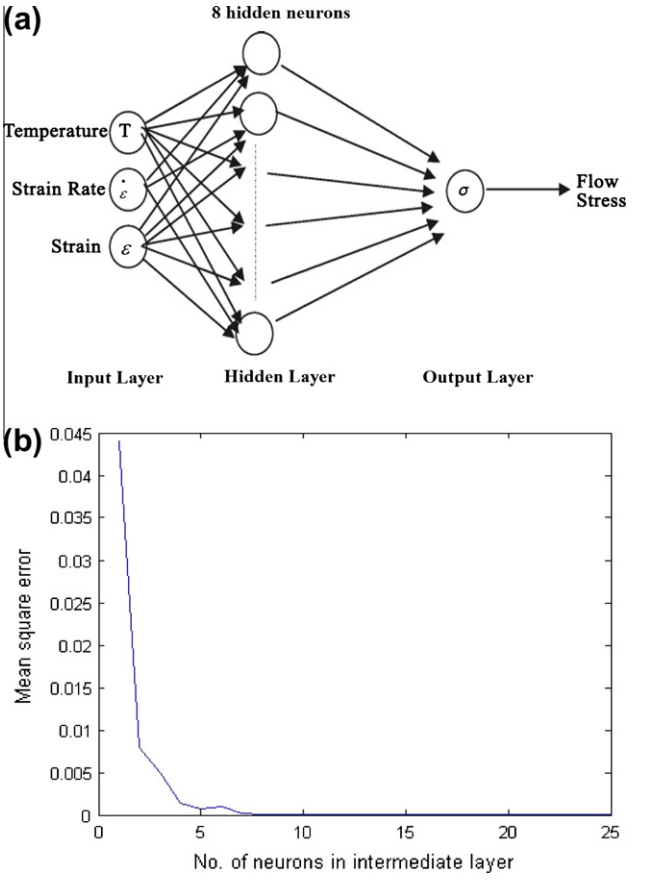


Fig. 4. (a) ANN architecture (3-8-1), (b) plot of mean square error (MSE) vs. no. of neurons in the intermediate layer of ANN.

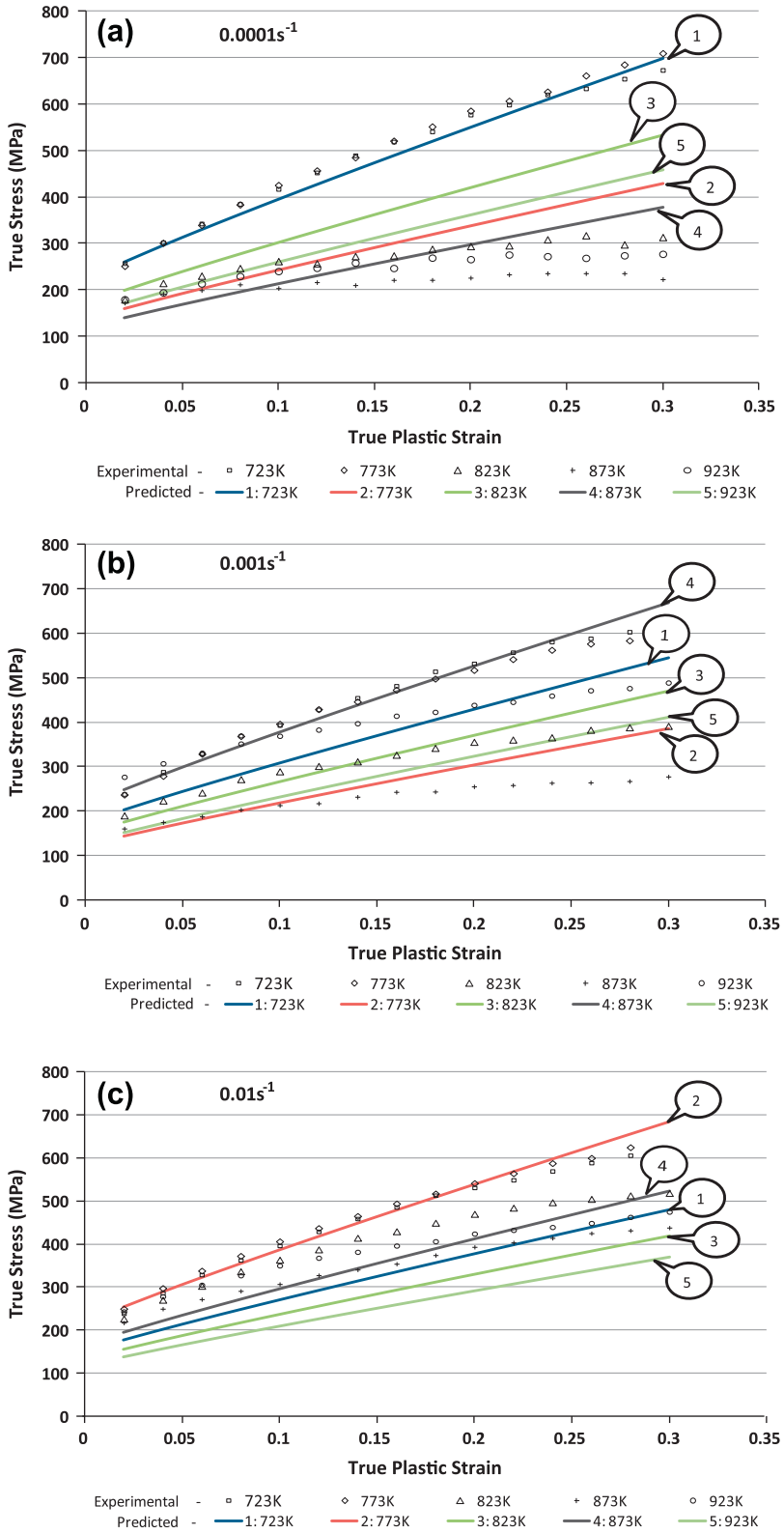


Fig. 5. Graphs showing experimental and predicted stress for JC model at various strains for strain rates of: (a) 0.0001 s^{-1}, (b) 0.001 s^{-1}, and (c) 0.01 s^{-1}.

exponential function. Based on the experimental data, the material constants of the model were identified by application of non-linear regression method. On the other hand, ANN model is a black box model, which considers the flow stress as an unknown function of strain, strain rate and temperature and adopts the training behavior.

3.1. Johnson Cook (JC) model

According to JC model [8,9], the flow stress is expressed as:

$$\sigma = (A + B\varepsilon^n)(1 + C\ln\dot{\varepsilon}^*)(1 - T^{*m}) \quad (1)$$

where σ stands for flow stress, A stands for yield stress at reference strain rate ($\dot{\varepsilon}_0$) and reference temperature (T_{ref}), B for coefficient of strain hardening, ε for plastic strain, $\dot{\varepsilon}^*$ for dimensionless strain rate (where $\dot{\varepsilon}^* = \dot{\varepsilon}/\dot{\varepsilon}_0$ with $\dot{\varepsilon}$ is strain rate) and T^* for homologous temperature (where $T^* = \frac{T - T_{ref}}{T_m - T_{ref}}$, with T as current absolute temperature, T_m as melting temperature and T_{ref} as reference temperature). For ASS304, the melting temperature is 1673 K, reference temperature is minimum temperature among all experiments (i.e. 723 K) and reference strain rate ($\dot{\varepsilon}_0$) is 0.01 s^{-1}. First, second and third terms in parenthesis in (1) refer to isotropic strain hardening, strain rate hardening and thermal softening. Material constants, n, C and m represent strain hardening exponent, coefficient of strain rate hardening and thermal softening exponent respectively. JC model considers all these three phenomena independently and their total effect is obtained by multiplying them.

At $T = T_{ref}$ and $\dot{\varepsilon} = \dot{\varepsilon}_0$, there is no strain rate hardening and thermal softening term. Hence, (1) reduces to: $\sigma = (A + B\varepsilon^n)$. Substituting A = 209.696 MPa (as calculated from the flow curve at 723 K and 0.01 s^{-1}) and taking logarithm on the two sides of the equation and then plotting $\ln(\sigma - A)$ vs. $\ln\varepsilon$ gives n = 0.9151 and B = 1383.2 MPa as calculated from its slope and intercept respectively. At $T = T_{ref}$, there is no thermal softening term in (1) (as $T^* = 0$) and it reduces to: $\sigma = (A + B\varepsilon^n)(1 + C\ln\dot{\varepsilon}^*)$. Then $\left[\frac{\sigma}{(A+B\varepsilon^n)} - 1\right]$ vs. $\ln\dot{\varepsilon}^*$ is plotted using flow stress data for every strain value at various strain rates. So, 15 different values of C are obtained from the slopes of the plots corresponding to 15 different values of strain. In the same way, taking $\dot{\varepsilon}_0 = \dot{\varepsilon}$, the strain rate hardening term vanishes from (1) (as $\dot{\varepsilon}^* = 1$), and it can be expressed as $\sigma = (A + B\varepsilon^n)(1 - T^{*m})$. So $\ln\left[1 - \left\{\frac{\sigma}{(A+B\varepsilon^n)}\right\}\right]$ vs. $\ln T^*$ is plotted using flow stress data for every strain value at different temperatures. Similarly 15 values of m are obtained from slopes of these graphs corresponding to 15 strain values.

Hence, total 15 values of C and m are obtained. The final values of C and m are obtained by method of least squares. A constrained optimization procedure is used to obtain the optimum values. This is done by reducing the average absolute error (Δ) between the experimental (σ_{exp}) and predicted flow stress (σ_p). The following equation gives us:

$$\Delta = \frac{1}{N}\sum_{i=1}^{i=N}\left|\frac{\sigma_{exp}^i - \sigma_p^i}{\sigma_{exp}^i}\right| \times 100 \quad (2)$$

where σ_{exp} is the experimental flow stress, σ_p is the predicted flow stress and N is the total number of data points being considered. The predictability of the constitutive equation can also be quantified by employing standard statistical parameters such as correlation coefficient (R). Correlation coefficient is a commonly used statistical tool which provides information on the strength of linear relationship between the experimental and predicted values. It is mathematically expressed as:

$$R = \frac{\sum_{i=1}^{i=N}\left(\sigma_{exp}^i - \bar{\sigma}_{exp}\right)\left(\sigma_p^i - \bar{\sigma}_p\right)}{\sqrt{\sum_{i=1}^{i=N}(\sigma_{exp}^i - \bar{\sigma}_{exp})^2 \sum_{i=1}^{i=N}(\sigma_p^i - \bar{\sigma}_p)^2}} \quad (3)$$

where $\bar{\sigma}_{exp}$ and $\bar{\sigma}_p$ are the average values of σ_{exp} and σ_p respectively. It is not necessary that the performance of the model is high if the value of R is high, as the model might have a tendency to be biased towards higher values or lower values of the data [17]. Hence, Δ which is computed through a term by term comparison of the relative error, is an unbiased statistics for measuring the predictability of the model [18].

The values of C and m obtained after constrained optimization are −0.0095 and 0.5417. All the constants obtained for JC model are listed in Table 3.

3.2. Modified Zerilli–Armstrong (ZA) Model

According to original ZA model [8,9], the flow stress is expressed as:

$$\sigma = A_0 + A_1\varepsilon^n \exp((-A_2T) + (A_3T\ln\dot{\varepsilon})) \quad (4)$$

where σ stands for flow stress, ε for equivalent plastic strain, $\dot{\varepsilon}$ for strain rate, T for absolute temperature, A_0 is the athermal component of yield stress, and A_1, A_2, A_3 and n are material parameters.

Flow stress is divided in two components by this model i.e. thermal and athermal. The thermal component is given by the term $A_1\varepsilon^n \exp((-A_2T) + (A_3T\ln\dot{\varepsilon}))$ while the athermal component which depends on grain size of deforming material and is denoted by A_0. It can be inferred from Fig. 3 that yield stress of ASS304 decreases with increase in temperature. So, in the m-ZA model, athermal component of flow stress is neglected and the modified equation for prediction of flow stress at elevated temperature is represented as:

$$\sigma = (C_1 + C_2\varepsilon^n)\exp\{-(C_3 + C_4\varepsilon)T^* + (C_5 + C_6T^*)\ln\dot{\varepsilon}^*\} \quad (5)$$

where σ, ε and $\dot{\varepsilon}^*$ are as used in (1) and $T^* = (T - T_{ref})$, where T is current temperature, T_{ref} is reference temperature (T_{ref} = 723 K as in JC model) and C_1, C_2, C_3, C_4, C_5, C_6, n are material constants. This m-ZA equation considers the phenomena of isotropic strain hardening, temperature softening, strain rate hardening, and the coupled effects of temperature, strain and of strain rate while predicting the flow stress at elevated temperatures.

At $\dot{\varepsilon}_0 = 0.01\ s^{-1}$, i.e. when $\dot{\varepsilon}^* = 1$, (5) can be expressed as:

$$\sigma = (C_1 + C_2\varepsilon^n)\exp\{-(C_3 + C_4\varepsilon)T^*\} \quad (6)$$

Taking natural logarithm on both sides of equation and plotting $\ln\sigma$ vs. T^* using flow stress data at 0.01 s^{-1} strain rate for each value of strain. The intercept (l_1) will give the value of $\ln(C_1 + C_2\varepsilon^n)$ and value of $-(C_3 + C_4\varepsilon)$ will be equal to slope (S_1) of the plot. C_1 is yield stress at reference strain rate ($\dot{\varepsilon}_0$) and reference temperature (T_{ref}) and is same as in JC model. Now, plotting $\ln(\exp l_1 - C_1)$ vs. $\ln\varepsilon$ after substituting C_1 = 209.696 MPa, the slope and intercept of the graph are n = 0.9231 and C_2 = 1463.1 MPa respectively.

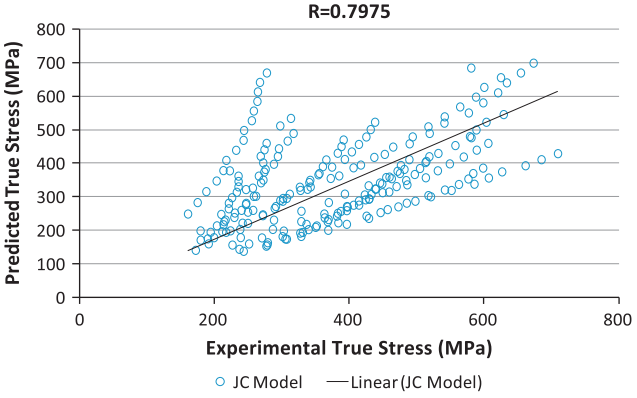


Fig. 6. Plot of predicted vs. experimental stress for JC Model.

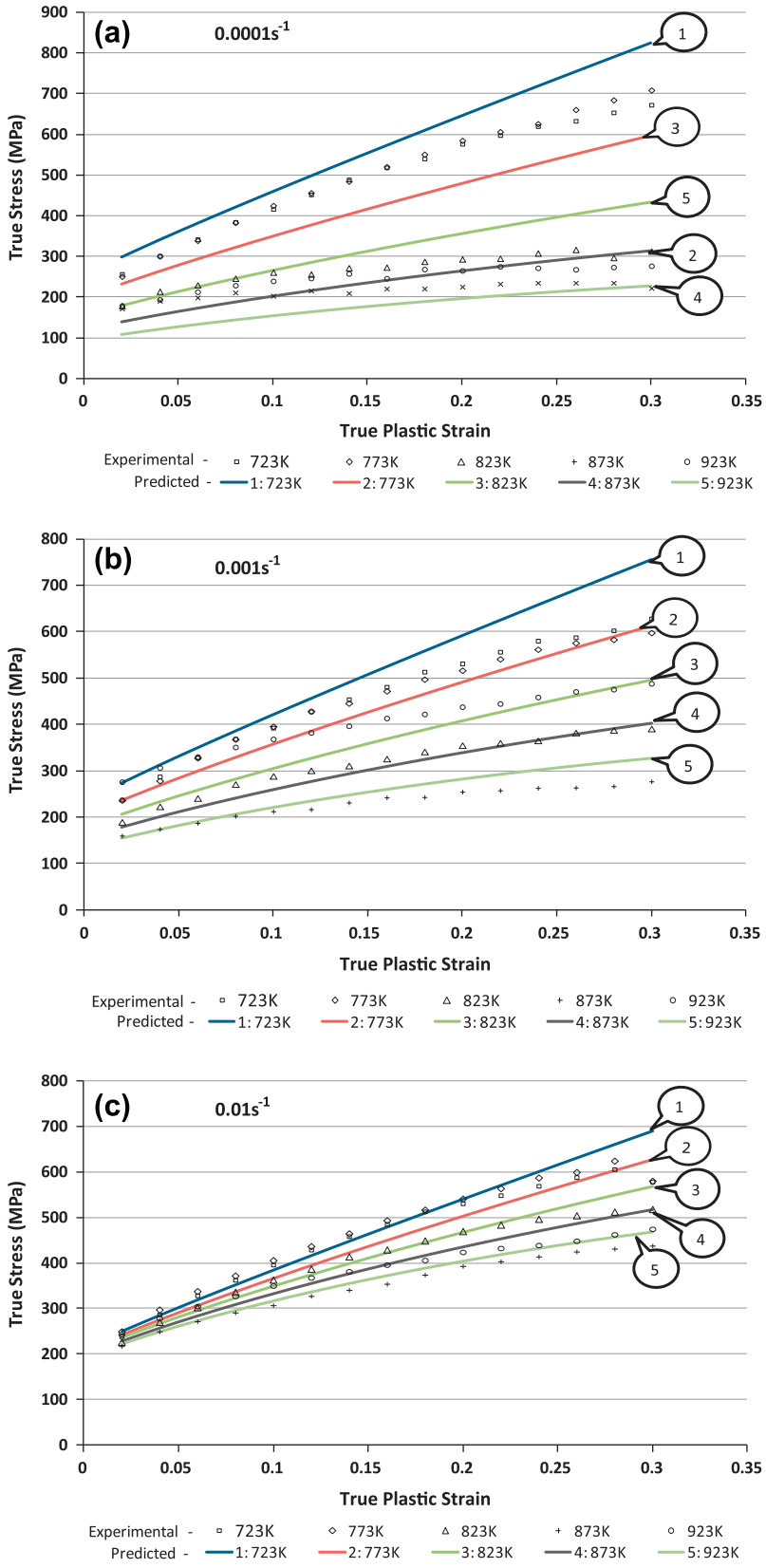


Fig. 7. Graphs showing experimental and predicted stress for modified ZA model at various strains for strain rates of: (a) 0.0001 s^{-1}, (b) 0.001 s^{-1} and (c) 0.01 s^{-1}.

Similarly plotting S_1 vs. ε, C_3 and C_4, can be calculated easily from slope and intercept of this graph. Value of C_3 and C_4 are listed in Table 4.

Now, taking logarithms, on both sides of (5) yields:

$$\ln\sigma = \ln(C_1 + C_2\varepsilon^n) - (C_3 + C_4\varepsilon)T^* + (C_5 + C_6T^*)\ln\dot{\varepsilon}^* \tag{7}$$

Slope (S_2) of $\ln\sigma$ vs. $\ln\dot{\varepsilon}^*$, gives the value of $(C_5 + C_6T^*)$. Total 75 different values of S_2 are obtained for five different temperatures at a particular strain and for total 15 strains. Now plotting S_2 vs. T^*, C_5 and C_6 are calculated from slope and intercept of this graph. Now total 15 values each of C_5 and C_6 are obtained for 75 different values of S_2 corresponding to five different temperatures. Material constants C_5 and C_6 are determined by least square method as mentioned in (2) and (3). The obtained values of C_5 and C_6 are −0.0386 and 0.0010 respectively. The computed material parameters for m-ZA model are presented in Table 4.

3.3. Modified – arrhenius type constitutive equation

Arrhenius – type equation [12] expresses the correlation between flow stresses, temperature and strain rate especially at elevated temperatures. Zener–Hollomon parameter in an exponential type equation represents the combined effects of temperature and strain rates on deformation behavior. These equations can be mathematically expressed as:

$$Z = \dot{\varepsilon}\exp\left(\frac{Q}{RT}\right) \tag{8}$$

$$\dot{\varepsilon} = AF(\sigma)\exp\left(-\frac{Q}{RT}\right) \tag{9}$$

where

$$F(\sigma) = \begin{cases} \sigma^n & \alpha\sigma < 0.8 \\ \exp(\beta\sigma) & \alpha\sigma > 1.2 \\ [\sinh(\alpha\sigma)]^n & \text{for all } \sigma \end{cases} \tag{10}$$

R is universal gas constant (8.314 J mol^{-1} K^{-1}), T is absolute temperature (K), σ is flow stress in MPa and Q is activation energy of hot deformation in kJ mol^{-1}. A, α and n are material constants, and $\alpha = \beta/n$ (see Ref. [12]).

Substituting $F(\sigma) = [\sinh(\alpha\sigma)]^n$ into (9) gives an ameliorated equation in a hyperbolic sinusoidal form, which describes dependency of steady–state flow stress on temperature and strain rate. Final equation after substitution is as follows:

$$\dot{\varepsilon} = A[\sinh(\alpha\sigma)]^n\exp\left(-\frac{Q}{RT}\right) \tag{11}$$

Flow stress can be written as a function of Zener–Hollomon parameter according to hyperbolic law.

$$\sigma = \frac{1}{\alpha}\ln\left\{\left(\frac{Z}{A}\right)^{1/n} + \left[\left(\frac{Z}{A}\right)^{2/n} + 1\right]^{1/2}\right\} \tag{12}$$

But the effect of strain is not taken into account by this constitutive equation. To consider this effect, Xiao and Guo [13] expressed the relationship between strain and flow stress as follows:

$$\sigma = \beta_0\varepsilon^{\beta_1}\exp(-\beta_2\varepsilon) \tag{13}$$

where β_0, β_1 and β_2 are constants.

Now, final constitutive equation which satisfactorily describe the effect of strain rate, temperature and strain on steady – state flow stresses, is developed by combining (12) and (13) as follows:

$$\sigma = \frac{\beta_0}{\alpha}\varepsilon^{\beta_1}\exp(-\beta_2\varepsilon)\ln\left\{\left(\frac{Z}{A}\right)^{1/n} + \left[\left(\frac{Z}{A}\right)^{2/n} + 1\right]^{1/2}\right\} \tag{14}$$

The material constants A, α, n, Q, β_0, β_1 and β_2 are determined by using the stress–strain data from the experiments done under different deformation temperatures and strain rates.

By substituting the power law and exponential law of $F(\sigma)$ into (9) following relationships are obtained for low stress levels ($\alpha\sigma < 0.8$) and high stress levels ($\alpha\sigma > 1.2$), respectively, $\dot{\varepsilon} = A_1\sigma^{n_1}$ and $\dot{\varepsilon} = A_2\exp(\beta\sigma)$ where A_1, A_2, and n_1 are material constants independent of experimental temperature. Taking logarithm on both sides of these equations and plotting $\ln\dot{\varepsilon}$ vs. $\ln\sigma$ and $\ln\dot{\varepsilon}$ vs. σ by substituting the values of flow stress and its corresponding strain rate under the strain of 0.02–0.3 and various temperatures, the relationships between flow stress and strain rate are obtained. n_1 and β can be calculated as mean slopes of the graphs. n_1 and β are 17.5471 and 0.02627 MPa^{-1} respectively, which gives value of $\alpha = \beta/n_1 = 0.0014969$ MPa^{-1}.

For all stress levels taking logarithm on both sides of (11), yields:

$$\ln[\sinh(\alpha\sigma)] = \frac{1}{n}\ln\dot{\varepsilon} + \frac{Q}{nRT} - \frac{1}{n}\ln A \tag{15}$$

The value of material constant n can be obtained as mean slopes of lines in $\ln[\sinh(\alpha\sigma)]$ vs. $\ln\dot{\varepsilon}$ plot, which is plotted by taking values of strain rate and flow stress at different strains and particular temperature and value of n was found to be 18.6278.

Q denotes activation energy and reflects the deformation difficulty degree in hot deformation. Now differentiating (15) at particular strain rate, yields:

$$Q = Rn\frac{d(\ln[\sinh(\alpha\sigma)])}{d(1/T)} \tag{16}$$

The value of Q, i.e. slope of $\ln[\sinh(\alpha\sigma)]$ vs. $1/T$ plot multiplied by $R * n$, is calculated for each strain and strain rate. Since the experiments have been conducted at 3 strain rates and 15 strain values are considered, which give a total 45 values of Q. Now, average of these Q values is 290.64 kJ mol^{-1}.

Zener–Hollomon parameter expresses effects of deformation temperature and strain rate on flow stress. Now substituting (9) into (11) and taking logarithm on two sides yields the following expression of Zener–Hollomon parameter:

$$\ln Z = \ln A + n\ln[\sinh(\alpha\sigma)] \tag{17}$$

Plot of $\ln[\sinh(\alpha\sigma)]$ vs. $\ln Z$ at a particular strain gives the value of $\ln A$. Now $\ln[\sinh(\alpha\sigma)]$ vs. $\ln Z$ is plotted for all 15 strain values and the average value of $\ln A$ is calculated as 42.1483. Thus, the value of A comes out to be 2.0173×10^{18} s^{-1}.

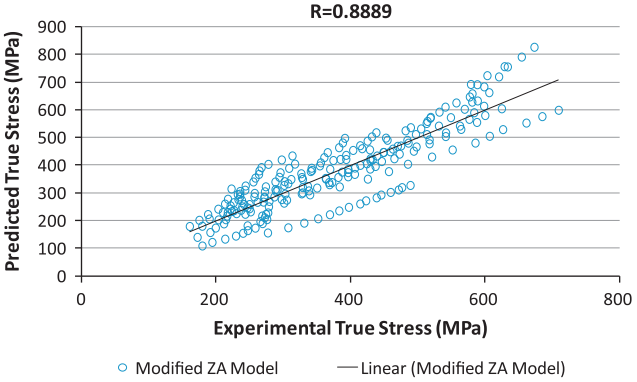


Fig. 8. Plot of predicted vs. experimental stress for modified-ZA model.

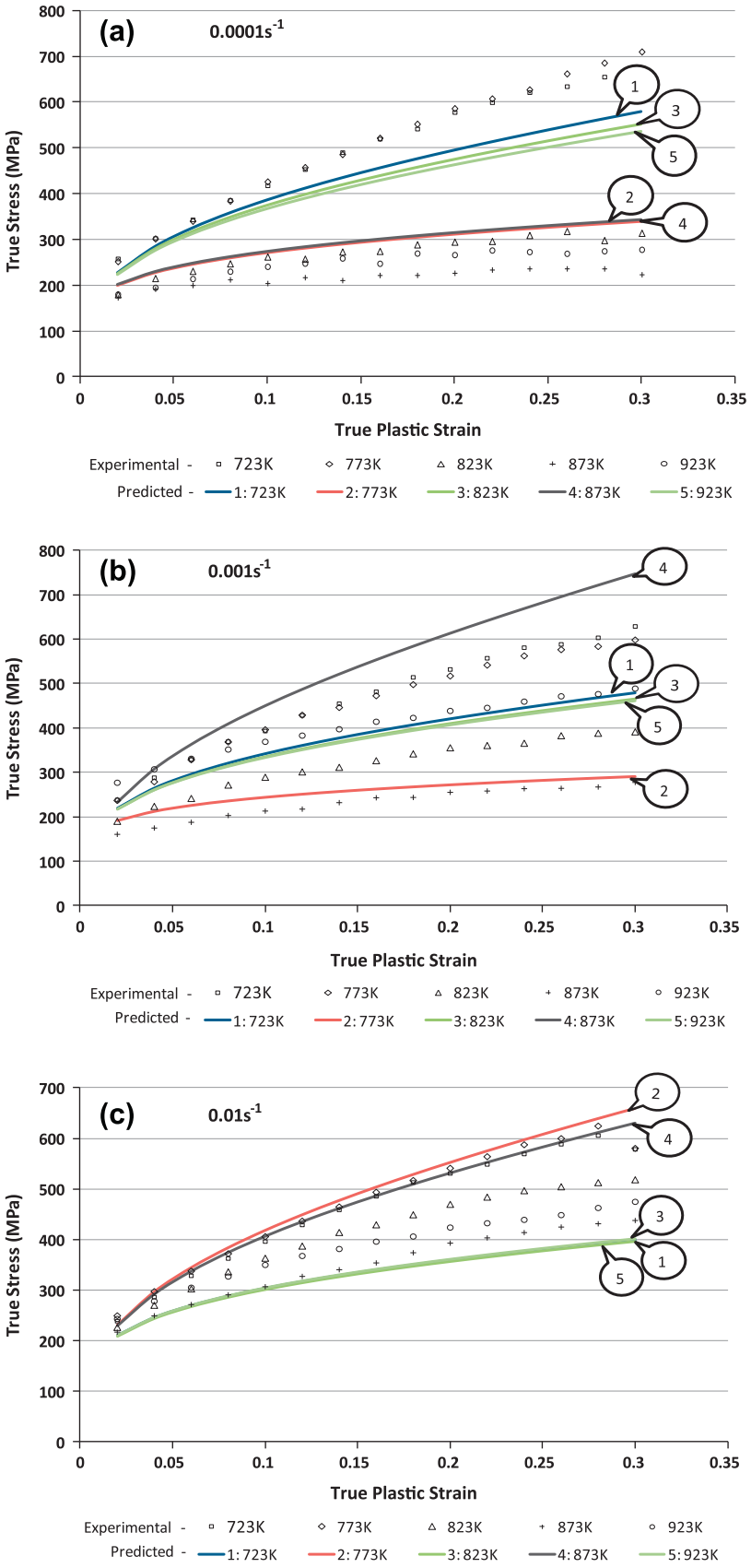


Fig. 9. Graphs showing experimental and predicted stress for modified Arrhenius model at various strains for strain rates of: (a) 0.0001 s^{-1}, (b) 0.001 s^{-1}, (c) 0.01 s^{-1}.

The material constants Q, A, n and α are listed in Table 5. Using these material constants the combined effect of deformation temperature and strain rate on flow stress at a particular strain can be described by following equation:

$$\sigma = \frac{1}{0.0015} \times \ln\left\{\left(\frac{Z}{2.01\times10^{18}}\right)^{1/18.6278} + \left[\left(\frac{Z}{2.01\times10^{18}}\right)^{2/18.6278} + 1\right]^{0.5}\right\} \quad (18)$$

where $Z = \dot{\varepsilon}\exp\left(\frac{290.64\times10^3}{8.314T}\right)$

Now, non-linear regression method as mentioned in [19] is used to determine the unknown parameters β_0, β_1 and β_2 called regression coefficients. Treating (14) as the fitting model, where the strain (ε) denotes independent variable, fitting model can be mathematically expressed as:

$$y = \frac{\beta_0}{\alpha}x^{\beta_1}\exp(-\beta_2 x)\ln\left\{\left(\frac{Z}{A}\right)^{1/n} + \left[\left(\frac{Z}{A}\right)^{2/n} + 1\right]^{1/2}\right\} \quad (19)$$

Denoting flow stress calculated from (18) by σ_z and putting it in (19) simplifies it as:

$$y = \beta_0 x^{\beta_1}\exp(-\beta_2 x)\sigma_z \quad (20)$$

Performing nonlinear regression analysis on experimental data obtained from 15 sets of experiments gives the values of constants β_0, β_1 and β_2 as shown in Tables 6–8 respectively. The values of β_0, β_1 and β_2 are not constant and change within a small extent due to change in strain rate and deformation temperature. Combined effects of temperature and strain rate on these constants can be represented by Zener–Holloman parameter. It is observed that these constants and $\ln Z$ follow a linear relationship as represented by following equations:

$$\left.\begin{aligned}\beta_0 &= 0.0395\cdot\ln Z - 0.0441\\ \beta_1 &= 0.0159\cdot\ln Z - 0.3088\\ \beta_2 &= -0.0216\cdot\ln Z + 0.5514\end{aligned}\right\} \quad (21)$$

3.4. Artificial Neural Networks (ANNs) model

ANN is a very powerful tool to correlate the input and output data for any system. It tries to simulate the neural networks structure of the human brain. The complex non-linear relationships between inputs and outputs are effectively captured and represented by it. Each neural network is composed of an input layer, an output layer and one or more hidden layers, which are connected by the processing units called neurons. Each neuron works as an independent processing element and has an associated transfer function. This transfer function describes how the weighted sum of its inputs is converted to the results into an output value. There are various training algorithms available to develop the ANN model. Of these, the back propagation learning algorithm is most popular in engineering applications. Back propagation algorithm is based on minimization of the quadratic cost function by tuning the network parameters. The mean square error (MSE) is considered as a measurement criterion for training set [20].

In this investigation, the flow stress is considered as a function of three parameters, i.e. strain, strain rate and temperature. Hence, these three parameters are the input variables of ANN and flow stress is the output variable. Hence, a feed forward network trained with the back propagation algorithm was developed, as shown in Fig. 4a. The next step in modeling is to choose the appropriate number of neurons in the intermediate layer to decide the ANN architecture. To check the ability of a particular architecture, the mean square error was calculated. The desirable network architecture should contain as minimal as possible hidden neurons and connections necessary for a good approximation of the true function. Fig. 4b presents the plot between the mean square error and the number of neurons in the intermediate layer. It can be observed that to avoid under-fitting as well as over-fitting of the data, the optimum number of neurons in the intermediate layer is chosen to be 8.

Before training the network, the input and output datasets have been normalized within the range of 0.05–0.95 to prevent a specific factor from dominating the learning for the ANN model. The main reason for normalizing the data matrix where the variables have been measured in different units is to recast them into the dimensionless units to remove the arbitrary effect of similarity between the objects. Thus, using (22) the experimental data was normalized to make the neural network training more efficient prior to the use of the datasets.

$$x_n = 0.05 + 0.90 * (x - x_{min})/(x_{max} - x_{min}) \quad (22)$$

where x_{min} and x_{max} are the minimum and maximum values of x, while x_n is the normalized data of the corresponding x. Once the best trained network is found, all the transformed data returns to their original value using following equation:

$$x = x_{min} + (x_n - 0.05) * (x_{max} - x_{min})/0.9 \quad (23)$$

4. Results and discussions

The equation for JC model is given by:

$$\sigma = (290.696 + 1383.2\varepsilon^{0.9151})(1 - 0.00951\ln\dot{\varepsilon})(1 - T^{*0.5147}) \quad (24)$$

Fig. 5a–c shows the predicted values of flow stress for JC model. As it can be seen from Table 3, the value of C obtained is negative. This indicates the negative strain rate sensitivity, which is responsible for the occurrence of DSA. However, the R value for the model came out to be 0.7975 as shown in Fig. 6 and value of average absolute error was 33.0266%. In addition, influence of the coupled effect of strain and temperature, and of strain rate and temperature needs to be considered. Hence, we can conclude that the flow behavior of ASS304 at elevated temperature is not governed only by three independent phenomena, viz., thermal softening, strain rate hardening and isotropic strain hardening. So, JC model is not so adequate in predicting the flow stress value in DSA regime. The correlation coefficient obtained in [8] was 0.977 and that in

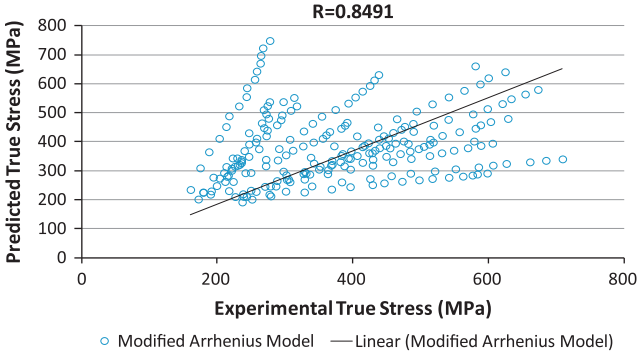


Fig. 10. Plot of predicted vs. experimental stress for modified Arrhenius model.

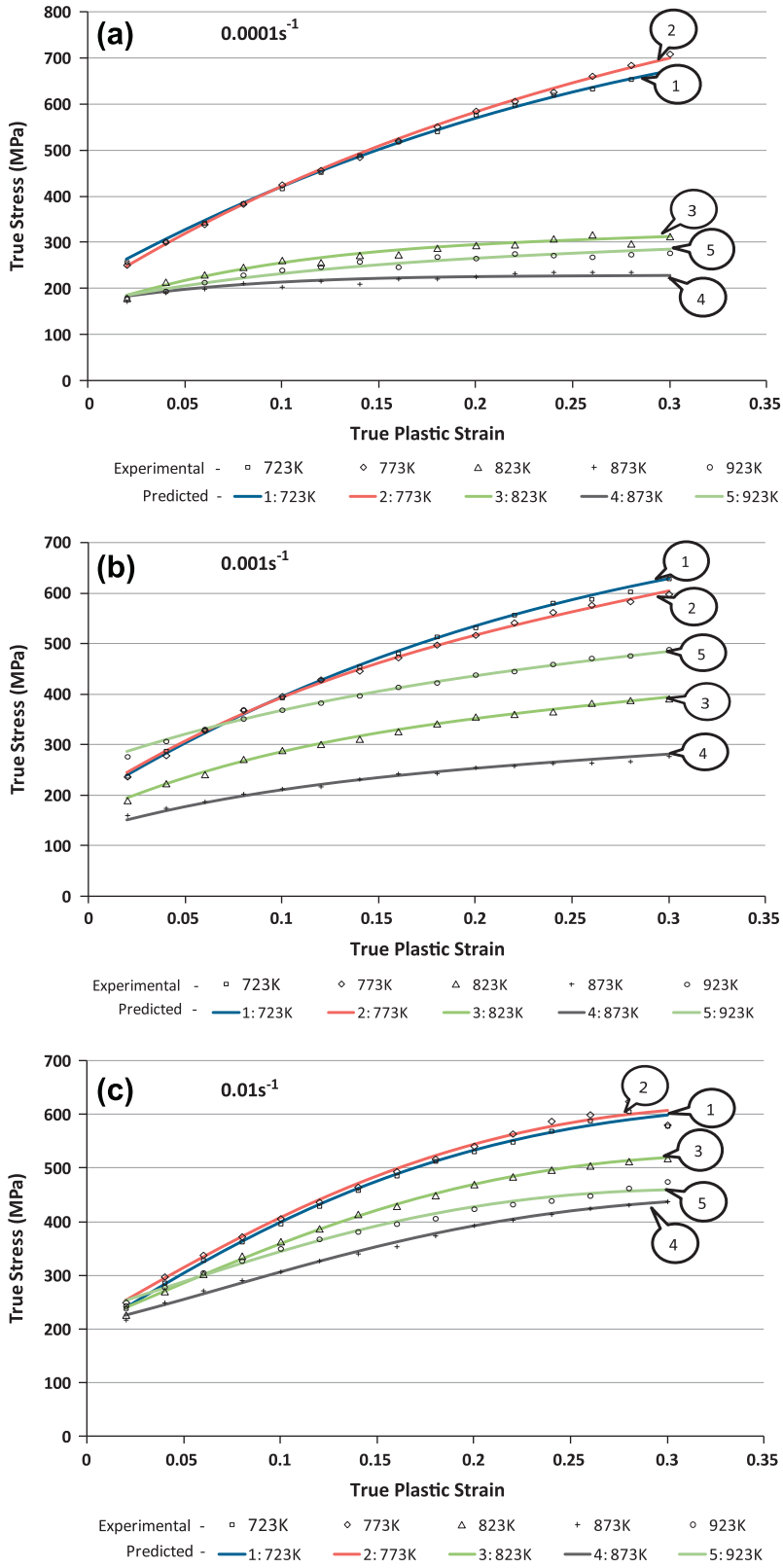


Fig. 11. Graphs showing experimental and predicted stress for ANN model at various strains for strain rates of: (a) 0.0001 s^{-1}, (b) 0.001 s^{-1}, (c) 0.01 s^{-1}.

[21] was 0.9423. This indicates that the prediction of JC model is more accurate only in the non-DSA regime, and not in the DSA regime.

The equation for modified-ZA model is given by:

$$\sigma = (209.696 + 1463.1\varepsilon^{0.9231}) \times \exp\{-(0.000475 + 0.0049\varepsilon)T^* + (-0.0386 + 0.0010)T^*)\ln\dot{\varepsilon}^*\} \quad (25)$$

Fig. 7a–c shows the predicted values of flow stress for m-ZA model. As it can be observed in Table 4 that the value of C_5 is negative, this indicates negative strain rate sensitivity. This is one of the major reasons for occurrence of DSA. For this model, the correlation coefficient R is determined to be 0.8889 as shown in Fig. 8 and Δ is 14.3916%. The m-ZA model is preferred over JC model as it considers the coupled effect of strain rate and temperature and also it captures the DSA phenomenon effectively. However, it is particularly not suitable to represent the flow behavior of material at temperatures above $0.6T_m$ and at lower strain rates [8]. For ASS304, the melting temperature is 1673 K. So, $0.6T_m$ = 1000 K. Although the experiments were carried out at temperatures lower than 1000 K, but the strain rates were low. Hence, m-ZA model was unable to predict the flow stress efficiently for the DSA regime. For the non-DSA regime, the predictions of ZA model were quite accurate as 0.992 in [8] and 0.9879 in [21]. This clearly shows that the ZA model is suitable for predicting the deformation behavior only in the non-DSA regime.

The equation for m-Arr model is given by:

$$\sigma = \frac{\beta_0\varepsilon^{\beta_1}\exp(-\beta_2\varepsilon)}{0.0015} \times \ln\left\{\left(\frac{Z}{2.01\times10^{18}}\right)^{1/18.6278} + \left[\left(\frac{Z}{2.01\times10^{18}}\right)^{2/18.6278} + 1\right]^{0.5}\right\} \quad (26)$$

where

$$\left.\begin{aligned}\beta_0 &= 0.0395\cdot\ln Z - 0.0441\\ \beta_1 &= 0.0159\cdot\ln Z - 0.3088\\ \beta_2 &= -0.0216\cdot\ln Z + 0.5514\end{aligned}\right\} \quad (27)$$

And Z is given by:

$$Z = \dot{\varepsilon}\exp\left(\frac{290.64\times10^3}{8.314T}\right) \quad (28)$$

Fig. 9a–c shows the plots between flow stress and true plastic strain values at different temperature and strain rates for m-Arr model. The correlation coefficient for this model is found to be 0.8491 as shown in Fig. 10 and Δ came out to be 15.1403%. The m-Arr model has been successful in predicting the flow stress of various alloys at elevated temperatures, but at higher strain rates [13]. In this investigation, elevated temperatures and low strain rates were considered, which falls under DSA regime. Upon analyzing the experimental data, it is found that this model is also not very effective in predicting the flow stress behavior in DSA regime. Xiao and Guo [13] reported that the average absolute error was less than 3.74% for 1Cr12Ni3Mo2VNbN martensitic steel .The error obtained when the model was used for ASS316 in the non-DSA regime [21] was 3.3432%. Even the correlation coefficient can be compared. Gupta et.al [21] reported the correlation coefficient to be 0.9852. Since the error value obtained for the DSA regime is 15.1403% and the correlation coefficient came out to be 0.8491, it can be said that modified Arrhenius model is also not effective in predicting the flow stress for the DSA regime.

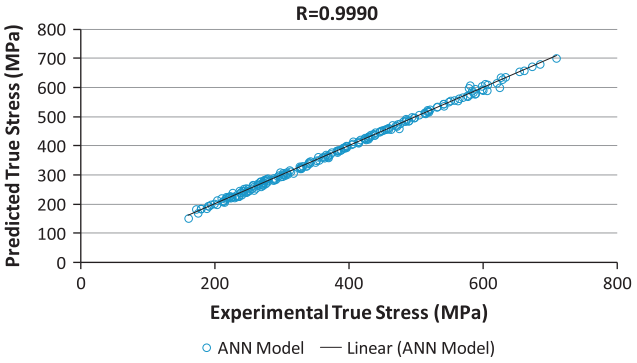


Fig. 12. Plot of predicted vs. experimental stress for ANN Model.

Table 9
Comparison between JC, m-ZA, m-Arrhenius and ANN models.

	JC	m-ZA	m-Arrhenius	ANN
R	0.7975	0.8889	0.8491	0.9990
Δ_{avg} (%)	33.0266	14.3916	15.1403	1.3746
Std. deviation (%)	15.3367	11.8737	12.4363	1.2293

For the ANN model, using 3-8-1 architecture, the flow stress values are predicted and presented in Fig. 11a–c, where the predicted values almost follow the same trend as the experimental values. ANN model gave a correlation coefficient of 0.9990 as shown in Fig. 12 and Δ value of 1.3746% which indicates an excellent correlation between experimental and predicted flow stress values. ANN model predicts data accurately even for the non-DSA regime, where the correlation coefficient obtained was 0.9983 [21]. When ANN was used to predict the deformation behavior of ASS316 in DSA regime the correlation coefficient obtained was 0.9955 [16]. This suggests that the ANN model possesses excellent capability to predict the flow softening stages and strain hardening, especially in the DSA regime. ANN model can be applied to serrated flow and mechanical properties can be predicted very accurately if the sufficient experimental input data is available in the DSA regime.

Table 9 presents the R value, Δ_{avg} and standard deviation of all the four models. From this table, it can be concluded that for predicting the flow stress of ASS304 in the DSA regime the ANN model is highly effective in comparison to the other three models, namely JC, m-ZA and m-Arr models.

5. Conclusion

A comparative study has been made to compare the capability of JC model, m-ZA model, m-Arr model and ANN model to predict the flow stress behavior of ASS304 in DSA regime. Based on the analysis done, the following conclusions can be made:

(1) ASS 304 exhibits phenomenon of DSA at elevated temperatures and lower strain rates as presented in Table 2.
(2) JC model is not suitable to provide flow stress predictions of ASS 304 in DSA regime as it does not consider coupled effects of strain, strain rate and temperature. While m-ZA and m-Arr models give better results than JC model, but they also could not capture DSA regime effectively and hence none of them provides an effective way of predicting flow stress in DSA regime.
(3) The correlation coefficient of ANN model is 0.9990, which clearly shows that it accurately reproduces the experimental flow stress in DSA regime. But, the present limitation of ANN

model is its integration with FEM software, where it can converse with the FEM software with the ability to take input from and give output to it. Hence, further studies are required either to develop this integration between ANN and FEM software or to develop some other constitutive models which can effectively predict the flow stress in DSA regime and can also be incorporated into the FEM software.

Acknowledgements

The financial support received for this research work from Department of Atomic Energy (DAE), Government of India, through Young Scientist Research Award 2009/36/45-BRNS/1751 is gratefully acknowledged.

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